List coverage periods and face amount patterns for term.

Coverage: 10–30 years or to age (e.g., 65)

Face Amount Patterns:

- **Level** (very common) – constant DB
- **Decreasing** – tied to declining balance (e.g., mortgage)
- **Increasing** – CPI-linked for inflation

List the 4 premium patterns for level term.

Source: ILA101-100: Ch. 1 – Term Insurance

1. **Level for entire period** (very common)
2. **Modified** – 1–2 increases over term
3. **Increasing every n years** – keeps initial premium/commission lower
4. **Level then ART**

Compare the 3 term premium schedules.

Source: ILA101-100: Ch. 1 – Term Insurance

1. **Attained age:**
   - Everyone age x pays same premium
   - Simplest to administer
   - Con: May not be competitive at all ages
2. **Select scale:**
   - Based on issue age + duration
   - Reflects UW effect
3. **Select & Ultimate (S&U):**
   - Grades from select $\rightarrow$ attained age
   - Assumes UW effects wear off
   - Pro: More competitive
   - Con: Future mortality $\uparrow$ risk

How to handle rising relative premiums in decreasing term?

Source: ILA101-100: Ch. 1 – Term Insurance

- Limited-pay premiums (may require CSV)
- Decreasing premium scales (less competitive)
- Floor DB (imperfect solution)
- OYT premiums (erratic, admin challenges)

Compare Par vs. Indeterminate term.

Source: ILA101-100: Ch. 1 – Term Insurance

1. **Par term:**
   - Higher premium $\rightarrow$ funds dividends
   - Pro: Higher persistency, cushion vs. adverse experience
   - Con: Expensive, not popular
2. **Indeterminate (non-guaranteed):**
   - Insurer adjusts premiums prospectively
   - Pro: Aggressive pricing
   - Con: Anti-selection

List other special term types.

Source: ILA101-100: Ch. 1 – Term Insurance

- Joint and second-to-die plans
- Hybrid term (blend of term + WL)
- Deposit term (pays endowment)

How does policy size affect premiums and lapses?

Premiums:

- Per \$1,000 ↓ as face ↑
- Reflects decreasing average costs
- Discounts larger than WL
- Highest bands → reinsured

Lapses:

- ↑ with policy size
- Different pattern than WL

Describe risk class and design-related premium differences.

Source: ILA101-100: Ch. 1 – Term Insurance

Risk Class:

- Smokers $>$ non-smokers
- Select risks → lower premiums

Gender: Sex-distinct or unisex rates

Design Practices:

- **Lapse-supported** – higher lapses → higher profit
- **Cross-subsidization** – price some cells aggressively

List key term and non-term riders.

Term Riders:

- Base insured, spouse, or child coverage
- One-Year Term (OYT) riders (purchase OYT with dividends)

Non-Term Riders:

1. **WOP** – Waives premium on qualified disability
2. **ROP** – Refunds premiums at term end or death
3. **ADB** – Accelerates DB on terminal illness
4. **Guaranteed Insurability** – Purchase more coverage without UW
5. **Disability** – Monthly income (% of original face)
6. **LTC** – Benefits independent of DB

Formula for Final DB when ADB utilized?

$$\text{Final DB} = \text{Original DB} - \text{Accelerated DBs}$$

What triggers ADB riders?

Chronic illness: Unable to perform 2/6 ADLs:

- Eating
- Bathing
- Dressing
- Toileting
- Transferring
- Continence

Critical illness: Heart attack, stroke, cancer, coronary bypass

How does term mortality compare to WL?

Most important pricing assumption

Aggregate term mortality $<$ **WL** because:

- Higher average policy size
- More select period exposure
- Conversions shift higher risks → WL

Summarize SOA high face Delphi study on term vs. WL mortality.

Source: ILA101-100: Ch. 1 – Term Insurance

- Term mortality $>$ **WL** (for high face amounts)
- Male improvement $>$ females, smokers
- Large policy mortality $<$ "all amounts"
- Best experience: **$1M+ band**

List key mortality considerations for post-level period and older ages.

Post-level period: Anti-selection (“conservation of deaths”)

Older ages: Credibility, large sizes, widow effect, select period wear-off

How do lapse rates impact mortality and profits?

Source: ILA101-100: Ch. 1 – Term Insurance

- **Anti-selection effects:** Lapses affect mortality
- **Impact by timing:**
  - **High early lapses** $\rightarrow$ harder to recover acquisition costs
  - **High late lapses** $\rightarrow$ erode profits
- **Premium structure:** Steep slopes $\uparrow$ lapses and anti-selection

Compare first-year vs. renewal term lapses to WL.

- **First year:** Term slightly $<$ **WL**
- **Renewal:** Term $>$ **WL**

Describe new trends in term underwriting.

Source: ILA101-100: Ch. 1 – Term Insurance

Traditional: Similar to WL

New trends:

- Select and super-select classes ↑
- AUW minimizes time, cost, intrusion
- Shift to **predictive analytics** vs. experience only

What are AUW characteristics and impact per SOA Delphi?

Source: ILA101-100: Ch. 1 – Term Insurance

- Uses **additional data sources** instead of fluids
- Sources: Rx, MIB, EHR
- Mortality $\uparrow$ **5–10%** (effect may diminish)

How do compensation and expenses affect term pricing vs. WL?

Source: ILA101-100: Ch. 1 – Term Insurance

Compensation:

- Term $<$ **WL** (lower premiums)
- Commissions sometimes "recycled" on renewal
- Agents may encourage lapses $\uparrow$ compensation

Expenses:

- Large % of premium (especially early ART)
- Term profits very sensitive to overhead allocation

Compare inflation risk: term vs. WL.

High inflation is **greater risk** for term than WL

Reason: WL has more assets backing larger reserves $\rightarrow$ cushion

List key regulatory aspects impacting term.

Source: ILA101-100: Ch. 1 – Term Insurance

- **Statutory reserves** – major profit impact
- **VM-20 (PBR)** replacing overly conservative methods (XXX)
- **State regulation:**
  - Coverage length (e.g., NY no coverage $>$ age 70)
  - Advertising, disclosure, marketing
  - Actuarial certifications
  - Contestibility period (typically 2 years)

List common term profit objectives with formulas.

Source: ILA101-100: Ch. 1 – Term Insurance

- **Profit Margin (PM):**

$$\text{PM} = \frac{\text{PV(Profits)}}{\text{PV(Premiums)}}$$

- **Return on Equity (ROE):**

$$\text{ROE} = \frac{\text{Net Income}}{\text{Equity}} \text{ (discounted or undiscounted)}$$

- **IRR:** Discount rate where $\text{PV(Profits)} = 0$
- **Surplus Strain:**

$$\text{Surplus Strain} = \frac{\text{First Year Income}}{\text{First Year Premium}}$$

- **BEY:** Year when accumulated assets $\geqslant$ reserves

What is term conversion and what are the benefits?

Source: ILA101-100: Ch. 1 – Term Insurance

- **Definition:** Policy converted from one type to another
- **Most Common:** Term → permanent (WL) with same UW rating
- **Timing:** Policyholder discretion, limited time frame, or automatic
- **Benefits:**
  - Increases sales (more attractive product)
  - Increases persistency (keeps healthy lives)
  - Permanent insurance more profitable than term

List methods to manage conversion costs.

- Terminal reserve $\rightarrow$ funds part of premium
- Pay commission only on discounted premium (varies)

List 3 approaches for charging conversion options.

1. **Charge only exercisers** $\rightarrow$ highest premium for converters
2. **Charge only those wanting option** $\rightarrow$ higher premium for option-holders
3. **Charge everyone**

List pricing assumptions for conversion options.

- **Election rate** – % of insureds electing
- **Coverage %** – Amount to be exercised
- **Lapse** – Low post-conversion
- **Mortality** – High when election rate low (anti-selection)

Formula for PV of extra mortality cost from conversions (age x, year r)?

Source: ILA101-100: Ch. 1 – Term Insurance

$$A_{x,r} = {}_{r}p_x \cdot e_{x,r} \cdot K_{x,r} \cdot v^r$$

where:

$$K_{x,r} = \sum_{t=1}^{\infty} ({}_{t-1}p_{x+r})(v^t)(q^C_{x+r+t-1} - q^S_{x+r+t-1})AR_{x+r+t}$$

- $A_{x,r}$: PV at age x of extra mortality cost (policy converted at year r)
- r: Duration since issue
- t: Duration since conversion
- $e_{x,r}$: Probability converts in duration r
- q^C: Mortality for converted WL
- q^S: Mortality for standard WL
- AR: Amount at risk each year

Formula for the cost of conversions per unit issued?

Source: ILA101-100: Ch. 1 – Term Insurance

Cost of conversions in year *t* per unit issued is the **sum of**:

1. The cost of expected conversions in year *t*

$$= \text{Reserve Credits} + \left(\frac{\text{Option Handling Expense}}{\text{Avg Units Issued Per Policy}}\right)\left(\frac{\text{Units Electing Option}}{\text{Radix}}\right)$$

2. The cumulative cost for years prior to *t*

$$= \sum\left(\frac{\text{Units Electing Options}}{\text{Radix}} \times \text{\% of Face Converted} \times \text{Option Charge}\right)$$

Radix = total number of units issued

List WL characteristics.

Source: ILA101-100: Ch. 2 – Whole Life

Permanent life insurance with:

- Coverage for life (if premiums paid)
- Tabular CSV
- Can be participating → dividends (cash, premiums, PUAs/OYT)
- **Premium patterns:**
    1. Ordinary level for life (most common)
    2. Indeterminate
    3. Limited payment

List WL pricing considerations.

Source: ILA101-100: Ch. 2 – Whole Life

Mortality (significant risk)

- No premium flexibility + very long coverage

Persistency (Lapse)

- Affects mortality and expense assumptions
- 2012 SOA study findings:
  - Lapse rates declining
  - Lapses $\uparrow$ in poor economy
  - Smaller policies $\rightarrow$ higher early lapses
  - Male $\approx$ female lapses
  - Stricter UW $\rightarrow$ lower early lapses

Underwriting – trending toward AUW like term

List UL characteristics.

Source: ILA101-100: Ch. 3 – Universal Life

1. Cash value: $\mathrm{CV}_t = (\mathrm{CV}_{t-1} + \mathrm{Prem}_t - \mathrm{Charges}_t)(1 + r)$
2. Credited rates: guaranteed min, portfolio vs. new money
   - Key risk: spread compression if rates $\downarrow$
3. Mortality charges: guaranteed max, attained age vs. S&U
4. Surrender charges common
5. Expenses: front-end loads, fixed charges
6. DB options: face or face + CV
7. Face amount changeable after issue
8. Flexible premiums
9. Partial withdrawals allowed
10. Policy loans allowed (lower credited rate)
11. Riders (similar to term)
12. Persistency bonuses

Describe fixed premium UL (FPUL).

Source: ILA101-100: Ch. 3 – Universal Life

- A.k.a. “interest-sensitive WL”
- Fixed premium like permanent products
- CV = max[Guaranteed CV, Net Accumulation Value]
- Vanishing premium design most popular

Describe ULSG:

1. Main designs
2. Reserve implications

1. **Minimum no-lapse premium (MNLP)** – most basic
   - SG based on cumulative minimum premiums paid
   - Generates little/no CSV
2. **Shadow account** – in force if shadow account > 0
   - Has own charges/credits (SA $\neq$ CSV!)

Reserve Implications: XXX (Reg 830) & AXXX (AG 38)

- AXXX greatly $\uparrow$ ULSG reserves
- Insurers needed relief $\rightarrow$ AG 48
- VM-20 replacing XXX/AXXX $\rightarrow$ simpler designs

How is interest credited in indexed UL?

$$\text{Index Crediting Rate} = \max\Big[\min(\text{Index Change} \times \text{Participation Rate}, \text{Cap}), \text{Floor}\Big]$$

Index change methods:

1. **Point-to-point** – index change between two points

$$\text{Index Change} = \frac{\text{Index at End of Segment}}{\text{Index at Beginning of Segment}} - 1$$

2. **Averaging** – average monthly index level

$$\text{Index Change} = \frac{\frac{1}{12}\sum_{m=1}^{12} \text{Index}_m}{\text{BOY Index}} - 1$$

List FPUL pros and cons.

Source: ILA101-100: Ch. 3 – Universal Life

- **Pros:**
  1. Similar to WL
  2. Higher commission
  3. Better persistency
- **Cons:**
  1. Lacks flexibility
  2. Admin complexity
  3. Vanishing premiums don't always vanish

List UL pricing considerations.

Source: ILA101-100: Ch. 3 – Universal Life

- Flexibility creates challenges (changing DBs, partial withdrawals, etc)
- Scenario testing important
- **UL profit sources:**
  1. Interest Earned – Interest Credited
  2. Cost of Insurance Charges – DBs Paid
  3. Expenses: Expense Charges + Surrender Charges – (Expenses + Commissions)
- Asset/liability analysis (interest rate risk)

Compare VUL vs. UL.

Source: ILA101-100: Ch. 4 – Variable Life Insurance

SIMILAR:

- Flexible or fixed premiums
- DB types: face or face + CV
- DB changes allowed
- Monthly charges (mortality, riders)
- Commissions
- Policy loan treatment

DIFFERENT:

- Premiums → separate account
- CSV varies with separate account performance
- No guaranteed CSV
- Sales loads more regulated

How does fixed premium variable life differ from WL?

Source: ILA101-100: Ch. 4 – Variable Life Insurance

- CSV uncertain, varies with separate account
- No guaranteed minimum
- DB varies by intervals (monthly or yearly)

Describe the Equitable design.

Source: ILA101-100: Ch. 4 – Variable Life Insurance

- Most popular in U.S.
- Fixed gross premiums
- Excess/negative performance buys positive/negative **paid-up additions**
- If S.A. outperforms AIR → DB continues rising

List survivorship insurance characteristics.

Source: ILA101-100: Ch. 5 – Survivorship Insurance

1. Joint and survivor – pays on last death
2. Wealth preservation
3. High face amounts
4. Par WL common – dividends buy PUA/term

List survivorship competitive measures.

Source: ILA101-100: Ch. 5 – Survivorship Insurance

1. Rate of return on death at specified duration
2. Minimum premium to fund benefits
3. Minimum premium to vanish in specified years
4. Minimum CSV to vanish premiums
5. Policyholders value DBs $>$ CSVs

List survivorship insurance riders.

Source: ILA101-100: Ch. 5 – Survivorship Insurance

- **PUAs and term riders** (very common)
- **Policy split rider**
  - Allows future policy split
  - Few sold, very valuable to some
- **Estate preservation rider**
  - Increases DB for estate taxes
- **First-to-die term rider** (estate planning)
  - Pays benefit when first dies
  - Uses: Pay-up policy, pay estate taxes, recover premiums

Compare single vs. dual status.

Source: ILA101-100: Ch. 5 – Survivorship Insurance

- **Single Status**
  - Blend of “one-alive and two-alive” status
  - Smoother CSVs and reserves
- **Dual Status**
  - 3 possibilities: Both alive, only X alive, only Y alive
  - CSVs and reserves jump after first death

List factors affecting single vs. dual status decision.

Source: ILA101-100: Ch. 5 – Survivorship Insurance

- Perceived marketability
- Administrative feasibility
- Regulators’ attitudes
- Perceived risk profile
- Implications of increased dual-status term rider costs

List 3 dual-life status calculation methods.

Source: ILA101-100: Ch. 5 – Survivorship Insurance

1. **Exact Age**
   - First principles using exact age mortality
   - Cumbersome: calculation, storage, validation
2. **Joint Equal Age**
   - Example: (55, 52) $\approx$ (53, 53)
3. **Equivalent Single Age**
   - Equates 2 ages to single age
   - Overcharges early $\rightarrow$ undercharges later

List 3 substandard rating methods.

1. **Age Rateup**
   - Assigns higher age to substandard insured
2. **Extra Premium**
   - Increase premium for substandards
3. **COI Multiple**
   - Increase UL COI by a multiple

List requirements for uninsurable lives.

Source: ILA101-100: Ch. 5 – Survivorship Insurance

1. Not terminally ill
2. Standalone underwriting required
3. Life expectancy $\geqslant$ 1-2 years
4. No increase in contagion factors
5. Not highly rated (Table D max)

List survivorship insurance pricing considerations.

Source: ILA101-100: Ch. 5 – Survivorship Insurance

- **Mortality**
  - Single life mortality of joint policyholders $\neq$ single life market
  - UW – concessions, medical
  - Contagion risk
  - Socio-economic class
  - Impact of very low lapses
- **Persistency**
  - Very low lapse rates common
- **Expenses**
  - Usually higher than single life business
- **Reinsurance**
  - Very important in pricing
  - Retention rates may be higher than other policies

List methods for determining extra mortality.

Source: ILA101-100: Ch. 6 – Substandard Life Insurance

- **Premium increase approaches:**
  1. **Flat extra:** extra amount per 1000
  2. **Table rating:** multiple of standard mortality
- **Determining extra mortality:**
  1. Numerical rating (sum of credits/debits) – arbitrary, may not properly evaluate risk
  2. Advance in age
  3. Lien method (increasing DB)
  4. Return of premiums
  5. Extra premiums + different nonforfeiture values
- **UL methods:**
  - Higher premium
  - Reduced coverage

How to subdivide substandard business?

1. **Male/female** (e.g. setback)
2. **Smoker/non-smoker** (e.g. “forgive” classes for non-smoker)
3. **By plan**
    - If Substandard/Standard ratio high → higher lapse, NT rates
    - Steeper slope → higher sales

List substandard expense considerations.

- **Allocation approaches:**
  1. Across all business
  2. Only to substandard
  3. Combination of 1 & 2 (most common)
- **Premium taxes and commissions:** Allocate directly to substandard
- **Acquisition expenses:** Higher than standard
- **Maintenance expenses:** Similar to standard

List substandard pricing considerations:

- Not-taken and lapse rates
- Premium paying period
- Ratings expiration
- CSV level vs. standard
- Profitability assessment

Source: ILA101-100: Ch. 6 – Substandard Life Insurance

- Not-taken and lapse rates higher than standard
- Limit premium paying period
- Ratings may be removed later
- CSVs slightly higher than standard
- Asset share tests fit premiums to profit objectives

Formula for extra substandard gross premium?

Source: ILA101-100: Ch. 6 – Substandard Life Insurance

1. Calculate standard GP:

$$\text{GP} = \text{NP}(1 + \text{ClaimCost\%}) + \frac{\text{AcqExp}}{\ddot{a}_{x:n}} + \text{OtherMaintExp} + \text{GP}(\text{\% Prem Exp})$$

2. Calculate substandard GP_R using same formula
   - Use substandard (rated) mortality and expense assumptions
3. Extra substandard GP:

$$\text{GP}_\text{R} - \text{GP}$$

How do acceleration riders work and what triggers them?

Source: Life Insurance Acceleration Riders

Insurer pays portion of face **before** death ("living benefits")

- Policyholder must meet trigger criteria

Trigger Events:

Terminal Illness	$\rightarrow$	Life expectancy $<$ 12-24 months
Chronic Illness	$\rightarrow$	Unable to perform 2+ ADLs (e.g. bathing)
Critical Illness	$\rightarrow$	Meet critical illness criteria (e.g. heart attack)

List 3 chronic illness benefit designs.

Source: Life Insurance Acceleration Riders

1. **Actuarial discounting**
   - Acceleration Benefit = PV of accelerated portion of face
2. **Lien method**
   - Lien = Acceleration benefit paid
   - Policyholder pays premium for full face
   - DB = Face – Lien
3. **Chronic Illness Rider**
   - Explicit additional premium

Tax treatment: Designed for favorable tax treatment (keep benefits tax-free)

List chronic illness acceleration risk controls.

Source: Life Insurance Acceleration Riders

1. Supplemental UW (screen for ADL loss, over-insureds)
2. Limit issue ages
3. Use lien or actuarial discounting
4. Limit acceleration amount (annual and max)
5. Require ADL certification (licensed health care practitioner)
6. Exclude temporary ADL losses
7. Explicit exclusions in trigger criteria
8. Limit to max table rating
9. Contestability follows base policy
10. Max benefit $<$ 100% of base DB

Describe reinsurance participation for acceleration riders.

Source: Life Insurance Acceleration Riders

Terminal Illness: Proportional to base policy participation

Chronic Illness:

Permanent products

- High probability of ultimate claim
- If chronically ill, lapse rate $\approx$ 0%
- Reinsurer participation: Full, limited to one-time payment, or none

Term products

- Much more uncertainty and risk
- Fewer ultimate death claims
- Life expectancy may be $>$ term
- Term conversions: additional risk

List deferred annuity product design features:

- Tax treatment
- Premium structure
- Charges
- Interest guarantees

Source: ILA101-101: Ch. 1 – Fixed Deferred Annuities

- **Tax treatment:** Qualified or non-qualified
- **SPDAs:** May have minimum required premiums (e.g. $5,000)
- **Charges:** FELs, periodic fees, SCs
- **FPDAs:** May have SCs based on premiums paid
- **Interest rates and guarantee periods:**
  - SPDAs: guarantee rate for 1–7 years
  - FPDAs: lower minimum guaranteed rates

List 4 deferred annuity types.

1. **CD Annuities**
   - SCs work like CDs
   - Penalty-free withdrawals 30–60 days after guarantee period (high lapses)
   - Low commissions
2. **Market Value Adjusted (MVA) Annuities**
   - Protects company from interest rate risk
   - AV $\downarrow$ as rates $\uparrow$
   - AV $\uparrow$ as rates $\downarrow$
3. **Two-Tiered Annuities**
   - 2 account values
   - Annuitization account rate $>$ surrender account rate
4. **Non-Surrenderable Annuities** (“Personal GICS”)

Formula for market value adjustments (MVA)?

Source: ILA101-101: Ch. 1 – Fixed Deferred Annuities

$$\text{MVA}_t = \left(\frac{1+a}{1+b+c}\right)^{n-t}$$

$$\begin{aligned} n &= \text{length of current guarantee period (years)} \\ t &= \text{years since guarantee period start} \\ a &= \text{current guaranteed rate} \\ b &= \text{current rate offered on similar product} \\ c &= \text{constant factor (0.000 to 0.005)} \end{aligned}$$

If SC applies:

$$\text{Final CSV}_t = \text{AV}_t \times \text{MVA}_t \times (1 - \text{SC\%}_t)$$

List deferred annuity features.

Source: ILA101-101: Ch. 1 – Fixed Deferred Annuities

1. Bailout provisions
2. Penalty-free withdrawal provision
    - Waives SC on portion of AV
    - May be constant % or % of premiums paid
    - May be tied to persistency
3. Return of principal guarantee provisions
4. Death benefits (usually DB = AV)
5. Waiver of SC on annuitization
6. Guaranteed settlement rates
7. AV enhancement bonuses: Annuitization, persistency, large AV, higher credited rate first year

Describe bailout provisions.

Source: ILA101-101: Ch. 1 – Fixed Deferred Annuities

- No SCs if Credited Rate $<$ Bailout Rate
- Medical bailouts (popular with age 50+ market)
- Total cost = option cost + additional surplus
- Option Cost $=$ Avg Lost SC $\times$ Excess Lapse Rate $\times$ Probability of Trigger
- Higher reserves and capital

List interest rate considerations.

Source: ILA101-101: Ch. 1 – Fixed Deferred Annuities

- **Interest Rate Risk** (a.k.a. C-3 risk, disintermediation)
- **Interest Spread** = Investment Earned Rate − Credited Rate
- **Target Spread Components:**
  1. Expenses (e.g. maintenance, commissions)
  2. Product features (e.g. bailouts)
  3. Risk charges (e.g. disintermediation)
  4. Expected profit
- **Crediting Strategies:**
  1. Ignore competition (fixed spread)
  2. Use competitor's rate as cap or floor
  3. Weighted average of #1 and competitor's rate

List deferred annuity pricing assumptions.

Source: ILA101-101: Ch. 1 – Fixed Deferred Annuities

1. **Withdrawal** (really important)
   - Affected by SCs, credited rates, distribution system, guarantees, policyholder characteristics
2. **Partial Withdrawal Provisions**
3. **Mortality** (less important)
4. **Commissions and Marketing Expenses**
   - Percent of premium
   - Affected by competitive pressures
   - Examples: SPDA 3–10%, FPDA 7% FY then 3%
5. **Expenses** (lower than life insurance)

List profit and pricing considerations for deferred annuities.

Source: ILA101-101: Ch. 1 – Fixed Deferred Annuities

- First-year strain sources: commissions, reserves, required capital
- Pricing should consider multiple interest rate scenarios
- Profit objectives: profit margin, IRR, break-even year, GAAP ROE
  - Make decisions considering all 3 together
  - Low IRR + high profit margin = surplus strain
- Pricing horizon usually 10–20 years

List variable annuity characteristics.

Source: ILA101-101: Ch. 2 – Variable Annuities

- Nearly all are SPDAs
- Passes C-3 risk to policyholder
- No guaranteed minimum CSV on separate account funds
- Basic VA DB = full AV (waive SCs)
- **Product charges:**
  - Similar to fixed annuities: FELs, SCs, periodic fees
  - % of asset charges: mortality, expense, profit, guarantee riders
- GMDBs and GLBs add substantial risk

List VA guarantees.

Source: ILA101-101: Ch. 2 – Variable Annuities

1. **GMDBs (Guaranteed minimum death benefits)** = max(AV, GMDB) on death
   - Step-up: GMDB = highest AV on any past anniversary $-\sum$ Withdrawals since
   - Roll-up: $\text{GMDB}_t = \text{GMDB}_{t-1} \times (1 + r) + \text{Premiums}_t - \text{Withdrawals}_t$
2. **GLBs (Guaranteed living benefits)** – must be alive to exercise
   - **GMIB:** Guaranteed minimum income benefit
   - **GMAB:** Guaranteed minimum accumulation benefits
   - **GMWB:** Guaranteed minimum withdrawal benefits
   - **GLWB:** GMWB for life (most popular)

Formulas for VA unit fund values?

$$\begin{aligned}
\text{Fund Value} &= \text{Units} \times \text{Unit Value} \\
\text{Units}_t &= \text{Units}_{t-1} + \text{Units Purchased}_t - \text{Units Withdrawn}_t \\
\text{Unit Value}_t &= \text{Unit Value}_{t-1} \times \text{NIF}_t
\end{aligned}$$

1. **Open-Ended Separate Account:**

$$\text{NIF}_t = 1 + \frac{\text{InvInc}_t + \text{UCG}_t + \text{RCG}_{t-1} - \text{Exp}_t}{\text{AV}_{t-1} + \text{Annuity Reserves}_{t-1}} - \text{Daily Asset Charges}$$

2. **Unit Investment Trusts:**

$$\text{NIF}_t = 1 + \frac{\text{Ending Value of Shares – Beginning Value of Shares}}{\text{AV}_{t-1} + \text{Annuity Reserves}_{t-1}}$$

Formula for VA annuitization into monthly payment?

$$\text{Pmt}_0 = \frac{\text{Account Value}}{12\,\ddot{a}_x^{(12)}}$$

$$\text{Pmt}_t = \text{Pmt}_{t-1} \times \frac{\text{NIF}_t}{(1 + \text{AIR})^{1/12}}$$

How to reduce variable payout variability?

1. Convert each annual payment to 1-year fixed annuity
2. Floor payment at 75% of initial payment
3. Floor payment at 100% and track actual payment separately
    - If actual payment $<$ floor, insurer “loans” shortfall
    - “Loans” repaid when actual payments $>$ floor
    - Outstanding loan usually forgiven on death

List VA pricing considerations (no guarantees).

Source: ILA101-101: Ch. 2 – Variable Annuities

1. Lapse rates (affect asset charge income)
   - Dynamic lapse assumptions critical
2. Premium persistency (difficult to estimate)
3. Average size (higher premium → higher profit)
4. Expenses (higher than fixed annuities)
5. Commissions (lower than life insurance)
6. Acquisition costs (recovery subject to equity risk)

List VA guarantee pricing considerations.

Source: ILA101-101: Ch. 2 – Variable Annuities

- GMDBs and GLBs require stochastic pricing
- Company must hold reserves for GMDBs and GLBs
- Pricing considerations to fund guarantee costs:
  - Future fund allocations assumption
  - Mean and variance of total returns
  - Mortality for GMDBs and GLWBs
  - GLB utilization (% electing and benefit amount elected)

List FIA characteristics.

FIA = <u>fixed</u> deferred annuity with GMAV and index credit potential

- GMAV $\geqslant$ minimum SNL value

Indexed Account Value (IAV):

- Credited interest based on index growth (e.g. S&P 500) floored at zero
- Index credits hedged with call options (or equivalents)

How is interest credited under point-to-point design?

Point-to-Point (PTP) – based on index growth over period

$$\text{IndexGrowth}_t = \frac{\text{Index}_t}{\text{Index}_{t-1}} - 1$$

Adjustments (most common to least):

$$\text{IndexCredit}_t = \begin{cases} \min\left(\text{IndexGrowth}_t, \text{Cap}\right) & \text{with cap} \\ \text{PartRate} \times \text{IndexGrowth}_t & \text{with participation rate} \\ \text{IndexGrowth}_t - \text{Fee\%} & \text{with \% of IAV fee} \end{cases}$$

Zero floor common: final index credit cannot be negative

How is interest credited under monthly sum cap design?

Average Monthly Sum – sum of monthly index growth during year

$$\text{IndexGrowth}_t = \sum_{i=1}^{12} \frac{\text{Index}_{t_i} - \text{Index}_{t_{i-1}}}{\text{Index}_{t_{i-1}}}$$

Monthly Sum Cap (common):

$$\text{IndexGrowth}_t = \sum_{i=1}^{12} \min\left(\text{MonthlyCap}, \frac{\text{Index}_{t_i} - \text{Index}_{t_{i-1}}}{\text{Index}_{t_{i-1}}}\right)$$

MSC typically not floored:

- Down months drag down return (unlike PTP)

List innovative or exotic FIA return methods.

- **Binary Returns** – e.g. index credit = 5% if index gains; else 0%
- **High Water** – based on highest index level

$$\text{IndexGrowth} = \frac{\text{Max Index Level}}{\text{Initial Index Level}}$$

- **Index Choice** – policyholder can choose their index (common)
- **Fixed interest account option**
- **Fund transfers on anniversaries**
- **Purchase inducements** – e.g. upfront bonuses or higher FY participation rates
- **GMAB** (a.k.a. “GMAV”) – guaranteed min value above SNL minimum
- **GMWBs** (cost less than VA GMWBs due to index credit floor)

How to price FIA from spread perspective? List risks.

Source: ILA101-101: Ch. 3 – Fixed Indexed Annuities

$$\text{Spread} = \text{Net Earned Rate} - \text{Pricing Spread} - \text{Hedging Budget}$$

- Pricing spread covers anticipated expenses and profit
- Hedging budget = option cost as % of fund value
- Adjust caps, participation, etc. to ensure option cost $\leqslant$ hedging budget

Profitability risks:

- Falling reinvestment rates
- High index credits $\rightarrow \uparrow$ IAVs $\rightarrow \uparrow$ option costs
- Higher future option costs caused by:
  - $\uparrow$ equity volatility
  - $\uparrow$ risk-free rates
  - Other factors

Describe FIA static hedging.

Static Hedging – uses OTC call spread options

- Call spread price = Call @ Current Index $-$ Call @ Cap

$$\text{Payoff}_t = \max(0, \text{Index}_t - \text{Lower Strike}) - \max(0, \text{Index}_t - \text{Upper Strike})$$

- **Funding ratio** = % of IAV hedged by options
  - Often $<$ 100% due to expected lapses
  - If actual lapses $>$ expected $\rightarrow$ insurer gains
  - If actual lapses $<$ expected $\rightarrow$ insurer loses profit

Describe FIA dynamic hedging.

Dynamic (Delta) Hedging – hold portfolio that replicates call spread option

- **Delta** = expected change in option value per unit change in index
- Main goal: hold Delta $\times$ notional amount of index
- Must be rebalanced regularly since Delta changes

Disadvantages:

- Involves "buying high and selling low"
- No downside protection like static hedging provides

How is GMAV funded for FIA? List key risks.

Source: ILA101-101: Ch. 3 – Fixed Indexed Annuities

- Funded with fixed interest bonds
- Key risks:
  - Bonds could default or lose value if rates ↑
  - Insurer may have to sell bonds early at loss to fund surrenders

Describe FIA reserving considerations.

FIA statutory reserving based on AG 33 and AG 35

- **AG 33** – prescribes CARVM for all fixed deferred annuities
  - Reserve = largest projected future CSV on PV basis
- **AG 35** – prescribes 4 AG 33-consistent methods specifically for FIAs
  - Type 1 method – HaR criteria must be met
  - Type 2 methods – HaR criteria do not have to be met
    - More complex and volatile than Type 1

List 5 income annuity types.

Source: ILA101-101: Ch. 4 – Income Annuities

1. **Payment Certain** ($a_{\overline{n|}}$)
2. **Specialty Markets**
   - Structured settlements, lottery winnings, gift annuities, reverse mortgages
3. **Life Contingent**
   - Life only (a_x)
   - Life with n years certain ($a_{\overline{n|}} + {}_{n|}a_x = a_{\overline{n|}} + {}_{n}p_x v^n a_{x+n}$)
   - Unit refund (Refund = CSV – Total Payments Received)
4. **Joint and Survivor** (a_{xy})
   - Pays until last death
   - Payment may change after first death
5. **Variable Payment** – adjusts payment for investment performance

List income annuity pricing considerations.

Source: ILA101-101: Ch. 4 – Income Annuities

1. Interest Rates (discount at spot rates)
2. Mortality (lower → higher PVFB)
   - Assume mortality improvement – e.g. 1% per year: $q_{x+t} \times (1 - 0.01)^t$
   - Substandard methods: Rated age (common), constant multiple, constant extra deaths
3. Expenses: premium taxes, maintenance expenses, commissions
4. Regulatory costs: taxes, cost of capital
5. Surplus strain impact on profitability
6. Cash flow pattern varies greatly by product type

Describe RILAs and compare with FIAs and VAs.

Source: Registered Index-Linked Annuities

- RILAs have characteristics of FIAs and VAs
  - Like FIA: Index credits hedged with options; subject to SNL
  - Like VA: AV can go down (maybe a lot)
  - RILA assets split between GA and SA
- Higher AV growth potential than FIAs (higher caps)
- Less downside protection than FIAs
  - **Floor** design: FIA with floor below 0%
  - **Buffer** design: insurer absorbs losses up to buffer
- Crediting terms: 1 to 6 years
- ROP riders common; GLWBs $\uparrow$ in popularity

Comparison of Deferred Annuities (Review back of card)

Source: Registered Index-Linked Annuities

Low Risk ⟷ **High Risk**

	FA	FIA	RILA	VA
General vs. Separate Account	GA	GA	Both	SA
Credited Interest	Declared	Indexed	Indexed	Market Return
Market losses possible	No	No	Yes	Yes
Contract values change daily	No	No	Yes	Yes
Subject to SNL	Yes	Yes	Yes	No
SC vs. MVA	Both	Both	Both	SC

List the 2 most common RILA crediting structures.

1. **Cap with floor:** Contractholder bears losses up to **floor**, then insurer bears all losses

$$\text{Credit} = \max\left[\text{Floor}, \min[\text{Cap}, \text{PartRate} \times \text{IndexReturn}]\right]$$

2. **Cap with buffer:** Insurer incurs losses up to **buffer**, then contractholder bears all losses

$$\text{Credit} = \begin{cases} \min\left[\text{Cap}, \min[0, \text{PartRate} \times \text{IndexReturn} + \text{Buffer}]\right] & \text{if IndexReturn} < 0 \\ \min\left[\text{Cap}, \text{PartRate} \times \text{IndexReturn}\right] & \text{otherwise} \end{cases}$$

Describe RILA rate setting and risk drivers.

Source: Registered Index-Linked Annuities

- **Rate setting process:**
  - Caps and buffers declared 1–2x/month
  - Example: 15% cap, 10% buffer, 1-year term on Nasdaq
- **Key rate drivers:** fixed income yields and option costs
  - Fixed income yields ↑ → option budgets ↑
  - Option costs vary with σ, risk-free rates, dividend yields, other factors
- **Rate sheet variable relationships:**
  - Caps ↑ as term ↑ (e.g. 1 or 6-year)
  - Buffer design: caps ↓ as buffer ↑
  - Buffer designs offer higher caps than floor designs
  - Caps vary by index (e.g. S&P 500, Nasdaq)

How is FIA option budget determined?

Source: Registered Index-Linked Annuities

FIA premiums invested in bonds of average credit rating (AA to BBB)

Insurer purchases call spread to hedge index credits

Option Budget = Earned Rate − Credit Default Charge − Expenses − Profit Margin

- **Credit default charge** = default losses on invested assets
- **Expenses:** commissions, acquisition, maintenance
- **Profit margin:** based on insurer's requirements and pricing guidelines
  - Aka "pricing spread" or "target spread"

How to calculate RILA hedging costs for buffer and floor designs?

Typical approach: purchase derivatives that replicate contract payoff

$$\textbf{Buffer}\text{ Design Option Cost} = \underbrace{\text{ATM Call} - \text{OTM Call}}_{\text{Call Spread}} - \text{OTM Put}$$

- Buy ATM call @ current index
- Sell OTM call @ cap
- Sell OTM put @ buffer

$$\textbf{Floor}\text{ Design Option Cost} = \underbrace{\text{ATM Call} - \text{OTM Call}}_{\text{Call Spread}} + \underbrace{\text{OTM Put} - \text{ATM Put}}_{\text{Put Spread}}$$

- Buy ATM call @ current index
- Sell OTM call @ cap
- Buy OTM put @ floor
- Sell ATM put @ current index

How are RILA interim values determined?

RILA values must be calculated daily (interim values)

- Available on surrender, death, etc.

Two main methods:

1. **Market value approach** – replicating option portfolio
   - Inputs = index return, interest rates, equity σ, dividend yield
   - Stronger match: minimizes balance sheet volatility and contractholder behavior risk
   - Easier to explain to regulators than pro-rata
2. **Pro-rata approach** – simple formula based on index return and time
   - Easier to explain to advisors/contractholders than MV approach
   - More balance sheet volatility and contractholder behavior risk

Describe RILA lock features.

Source: Registered Index-Linked Annuities

Some RILAs allow contractholder to lock in interim values

- **Manual lock** – contractholder can lock interim value on any non-anniversary day
  - Locks in AV for remainder of index term
  - Reduced only for withdrawals
- **Automatic lock** – contractholder can provide target value to lock in
  - Once interim value hits target, it is locked in <u>permanently</u>
  - Target values can be modified until lock takes place

Describe common RILA riders.

ROP on death ("ROP DB") is most common rider

- DB = max[Interim Value, Premiums Paid $-$ Withdrawals]

GLWB riders more popular in recent years

- Traditional benefit base: income payments = withdrawal % $\times$ benefit base
  - Benefit base = nominal amount that grows at specified rate independent of AV
  - Withdrawal % varies with age when withdrawals start
  - If withdrawal $>$ guaranteed amount $\rightarrow$ reduces future guaranteed withdrawals
- Non-traditional (no benefit base): initial income payment = % of AV

Describe RILA hedging risk management.

Source: Registered Index-Linked Annuities

- **Hedging with options** – major considerations:
  1. **Transaction size:** trading cost lowest when notional $\geqslant$ \$1M
     - Some insurers restrict index term start dates to 2–4x/mo to increase trade sizes
  2. **Counterparty risk:** option use introduces counterparty risk
  3. **Liquidity:** put options sold require collateral
     - Collateral calls typically met with cash or treasuries
  4. **Decrements:** hedge positions should reflect deaths, surrenders, withdrawals
- **Hedging with VA GMxBs instead of options**
  - VA GMxBs typically have opposite equity market exposure as RILAs
  - May be cheaper with less counterparty risk

Describe 4 RILA risks.

Source: Registered Index-Linked Annuities

- **Rate setting risk** – profit ↓ as option prices ↑ after rate declaration
  - Mitigate: more frequent rate adjustments
- **Investment risk** – market and default risk in fixed income assets
  - Mitigate: higher quality assets and appropriate pricing
- **Liquidity risk:** contractholder behavior and collateral calls
  - If surrenders spike → insurer may have to sell assets
  - If index falls enough → insurer may have to sell assets to meet collateral calls
- **Operational risk** – RILAs require significant expertise and complex operations
  - Require expertise: hedging, investments, ALM, contract design
  - Higher sophistication → higher risk of improper execution

Describe RILA asset management and regulatory considerations.

Source: Registered Index-Linked Annuities

- **General vs. separate account management**
  - GA assets often intermingled and not allocated to specific LOBs
  - SA assets more often allocated to specific product/contract
  - RILA assets held in both GA and SA
  - RILA practice still emerging
- **Reserves and regulations**
  - NAIC developing nonforfeiture standard for RILAs
  - RILAs follow VM-21 for US statutory reserves
  - RILA RBC requirements follow those for VAs
  - Under US GAAP:
    - Base RILA contract follows ASC 815
    - RILA riders classified as market risk benefits (MRBs)

Why do governments view structured settlements positively?

Source: Structured Settlement Annuities

- Helps control costs and reliance on government benefit programs
- Less likely to be squandered than lump sum
- Plaintiff doesn't have to invest/manage large sum

List key steps to create a structured settlement.

Source: Structured Settlement Annuities

- Defendant/casualty insurer agrees to pay settlement
- Defendant purchases annuity contract from life insurer
  - Marketed by specialized field force
  - Consultants work between defendant and life insurer (earn commission)
- Nearly all US structured settlements use qualified assignment
  - **Qualified assignment** – defendant pays assignee to take over liability
  - Assignee usually subsidiary of life insurer
  - Releases defendant/casualty insurer from liability
- Life insurer makes periodic payments to plaintiff

List benefit payment forms for structured settlements.

Source: Structured Settlement Annuities

1. **Annuity certain** – specified payments at periodic intervals over predetermined years
2. **Single life annuity** – specified payments at periodic intervals for life
    - Certain and life is common
3. **Temporary life annuity** – single life annuity ending at specified date or death
4. **Lump sum** – single payment at specified future date
    - Can be certain or life-contingent
    - Settlements often call for series of lump sums (e.g. every 5 years)

Benefit escalator – payments increase at fixed rate (e.g. 2%) to protect against inflation

Describe mortality pricing considerations for structured settlements.

Source: Structured Settlement Annuities

Mortality assumption – applicable to life-contingent payments

- UW challenging
- Settlement population $\neq$ normal insured populations
- Settlement annuitant characteristics:
  - Younger (before retirement)
  - No anti-selection
  - Low credibility (very limited data)
  - Often have significant trauma (e.g. spinal injury) – very different from “normal” substandard cases (obesity, diabetes)
- **Age rate-ups** common: set higher issue age
- **Mortality improvement:** usually based on 1D scale (e.g. only differs by gender)

Describe interest rate pricing considerations for structured settlements.

Source: Structured Settlement Annuities

Interest rates – required to discount benefit payments

- Structured settlements are **long-tailed liabilities** (can extend well beyond 30 years)
- Max bond length usually 30 years (Treasuries, corporates)
- Fixed nature of benefits allows investment in shorter, less liquid assets
  - Private placements, mortgage-backed, etc.
- Final rates = gross yields $-$ margins for defaults, investment expense, profit

How are expense assumptions set for structured settlements?

Source: Structured Settlement Annuities

- **Percent of premium assumptions:**
  - Commissions and acquisition expenses that vary with premium
  - Premium taxes
- **Fixed fee assumptions:** good for large contracts; bad for small contracts
  - Acquisition costs that don't vary with premium
- **Percent of benefit assumptions:**
  - Maintenance and overhead

Describe pricing considerations:

- Post-tax return on invested surplus
- Administration of pricing/rate changes

Source: Structured Settlement Annuities

- **Post-Tax Return on Invested Surplus:**
  - Surplus strain from RC and conservative stat reserves
  - Higher surplus strain → lower IRR
  - Account for federal and state income taxes
- **Administration of Pricing/Rate Changes:**
  - Ratebook guarantee – insurers usually guarantee annuity rates for period
  - Hold additional interest rate margin to hedge risk of falling rates

Describe mortality risk considerations for structured settlements.

Source: Structured Settlement Annuities

- Credibility methods require expected basis: hard to find
- Challenges using SOA studies:
  - Tend to cover much longer periods
  - Substandard SOA experience covers wide range of impairments
  - UW standards may have changed over time
- Anti-selection doesn't occur in early durations as with life insurance
- Very high competition amplifies bad UW risk
- Mortality improvement could be substantial as medical treatment advances
  - E.g. incurable spinal injury could be cured in future

Describe interest rate and A/L mismatch risk considerations for structured settlements.

Source: Structured Settlement Annuities

- **Typical approach:**
  - Match A/L cash flows for limited years
  - Duration match in total
- **Substantial reinvestment risk:** liability duration $>$ asset duration
- Strategies to mitigate reinvestment risk:
  - Derivatives (futures, swaps)
  - Long-maturity zero-coupon bonds
  - Equities (but these increase RC)
- Callable bonds and MBS introduce **prepayment risk**
- To hedge **ratebook guarantee risk**, impose dollar limits on published rates

How do benefit escalators impact structured settlement risks?

Source: Structured Settlement Annuities

Benefit escalators magnify mortality, interest rate, and reinvestment risk

- Falling rates + mortality improvement = bad combo
- Increasing payments offset discounting effect
- Increases reinvestment risk beyond investment horizon

Describe reserving and valuation for structured settlements.

Source: Structured Settlement Annuities

- **Statutory Reserving** – based on VM-22 for payout annuities
  - Follows VM-22 for payouts:
    - Reserve = PV of payments using prescribed mortality and interest
    - May be modified to PBR approach in future (CTE 70)
  - AG IX-A (9-A) prescribes "constant extra death method" for substandard cases
    - Add deaths per 1000 until life expectancy matches UW
    - Results in reserves that grade to standard basis
    - Tends to produce mortality rates too high early, then too low later
  - AG IX-B (9-B) provides guidance on durational interest rates
    - Assigns new rate if benefit amounts exceed 110% of prior year
  - Structured settlements subject to asset adequacy testing in US
- **GAAP Reserving** – follows usual payout approach using rated age

Describe for structured settlements:

- Annuitization of attorneys' fees
- Factoring
- Inflation protection variations

Source: Structured Settlement Annuities

- **Attorneys' fees paid as annuity:**
  - Pricing similar to typical retirement annuities
  - Not tax-free (not structured settlements)
- **Factoring** (US only) – annuitant sells some/all payments to third party
  - Exists because no other way to cash surrender contract
  - Controversial: price may use high discount rate
- **Inflation protection variations:**
  - Most popular: benefit escalators
  - Some indexed to CPI
  - "Market indexed" to equity index (e.g. S&P 500) with guaranteed floor

Describe unique aspects of Canadian structured settlement market.

Source: Structured Settlement Annuities

- Canadian structured settlements also tax-free
  - Canadian tax authority = Canada Revenue Agency (CRA)
- Notable differences with US market:
  - 50% of structured settlements involve qualified assignments ($\sim$100% in US)
  - Market much smaller (2–3 insurers dominate)
  - No factoring since structured settlement payments are non-transferrable

Define 2 types of group annuities for pension sponsors.

Source: Pension Risk Transfer in Canada and the US

1. **Buy-out annuity** – insurer pays pension directly to plan members
2. **Buy-in annuity** – insurer makes monthly bulk payment to pension fund, which pays members

List advantages of each type of group annuity.

Source: Pension Risk Transfer in Canada and the US

Group annuity purchases benefit all stakeholders

Advantage	Buy-Out	Buy-In
Reduces sponsor's risk of future cash contributions	✓	✓
Benefits paid by well-regulated insurers	✓	✓
Avoids undesirable upfront cash and accounting impacts		✓
Eliminates sponsor's operational risk in making payments	✓	
Eliminates PBGC premiums for sponsor	✓	
Can be converted to the other type		✓

Compare US vs. Canada group annuity markets.

Source: Pension Risk Transfer in Canada and the US

Both markets $\approx$ tripled 2016-2021

- US: $38.1bn, 19 insurers
- Canada: $7.7bn, 8 insurers

Lift-out (US term) – group annuity used for de-risking (more common than winding-up)

Insurers may specialize by:

- Buy-out vs. buy-in
- Small vs. large transactions
- Retired vs. deferred annuitants
- CPI-linked vs. nominal benefits
- Longevity profiles

Outline steps in group annuity purchase process.

Source: Pension Risk Transfer in Canada and the US

1. **Sponsor engages consultant for bids**
2. **Consultant requests quotes** (members covered, annuity type, tranching, key dates)
3. **Insurers submit quotes**
4. **Sponsor selects insurer based on:** price, fiduciary duty, insurer ratings/experience
5. **Weeks pass before premium payment** → exposes insurer to market risk
    - Hedged via premium roll-forward or in-kind asset transfer

Describe group annuity policy differences from individual annuity.

Source: Pension Risk Transfer in Canada and the US

- Group contract = insurer + plan sponsor
  - Buy-outs only: annuitants receive certificate of benefit entitlement
- Buy-out annuities mirror pension plan benefits
  - Benefits may be CPI-linked or integrated with government benefits
- Buy-in annuities may pay level benefit, sponsor covers inflation increases

Outline policy requirements for deferred annuitants.

Source: Pension Risk Transfer in Canada and the US

- Commutation options
- Retirement options:
  - Allowed retirement date
  - Penalties for early retirement
  - Pension forms available
- Buy-ins allow cashouts in specific circumstances per regulations

List main assumptions for group annuity pricing.

Source: Pension Risk Transfer in Canada and the US

1. Expected asset return
2. Expected asset defaults
3. Longevity base table and improvement scale
4. Acquisition and maintenance expenses
5. Annuitant behavior
6. Provisions for adverse deviation
7. Profitability level

Outline pricing considerations for group annuities.

Source: Pension Risk Transfer in Canada and the US

1. **Asset mix** – heavily influences pricing, can change daily
    - Group annuity CFs mostly illiquid $\rightarrow$ allows investment in illiquid assets
    - Commutation options $\uparrow$ liquidity needs
2. **Longevity** – assumptions vary widely by insurer
    - Pricing depends on date longevity risk transfers from sponsor to insurer
    - Assumptions based on: zip code, job type, experience data
    - Data consistency allows more aggressive pricing
    - Mortality improvement usually based on industry tables
3. **Liquidity** – relevant if deferred annuitants allowed to commute
    - Requires assumption: % expected to choose commutation
4. **Market risk margins** – for adverse market movements after quote
    - Reduced by premium roll-forward or in-kind asset transfer
5. **Data risk margins** – based on insurer's ability to adjust/reprice
6. **Annuitant behavior** – relevant if deferred annuitants have options
    - Subsidized options require assumptions for retirement date and pension form
7. **Consultant commissions** – usually paid by sponsor or pension fund
8. **Reinsurance or offshore structures used by insurer**

Outline risk mitigation approaches for group annuities.

Source: Pension Risk Transfer in Canada and the US

- **Investment risk:**
  - Careful credit UW
  - Invest in higher quality fixed income assets
  - Limit non-fixed income investments
- **Interest rate risk:**
  - Closely match A/L duration at key rates
- **Longevity risk:** challenging (lifestyle, environment, medicine, pandemics)
  - Limit # deferred annuitants
  - Use conservative assumptions if data points misaligned
  - Issue life business for natural hedge
  - Reinsure longevity risk
- **Annuitant behavior risk:**
  - Use conservative assumptions for retirement date and pension form
- **Operational risk** – incorrect payments, paying deceased annuitants
  - Mitigate by regularly verifying annuitants alive